KLH UNIVERSITY
HYDERABAD



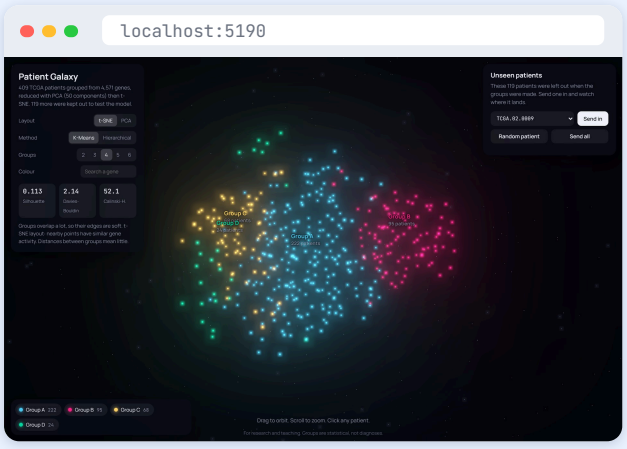
RESULTS: FRONT END

Team Number
T.N. – 10

Team Members
2520030184 **Aniketh Pani**
2520030187 **Harshit Sajjanapu**

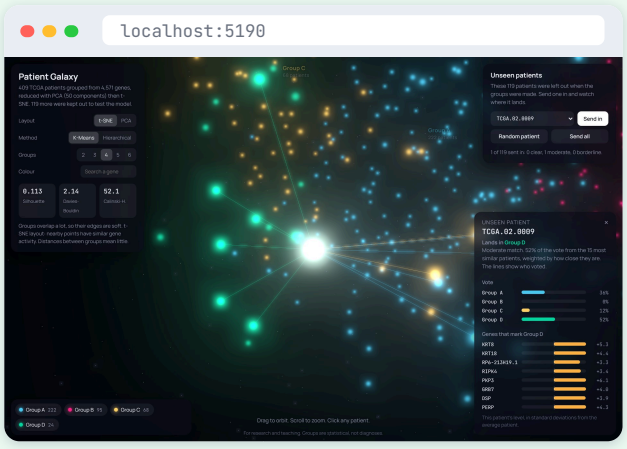
Guide Name
Mr. Shaik Asif

1. Patient Galaxy (Home Page)



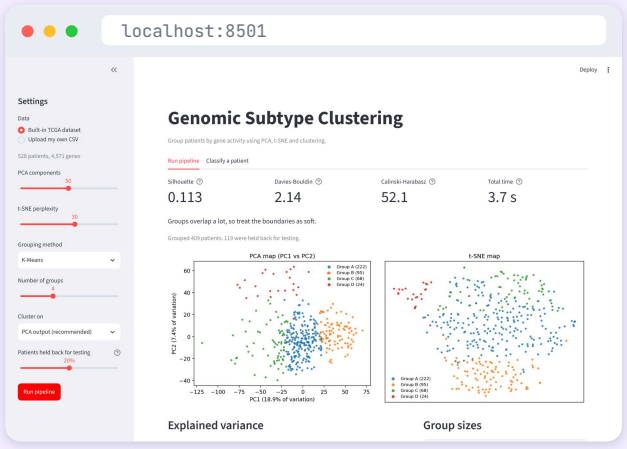
409 TCGA patients as glowing points in 3D, coloured by K-Means group. Drag to orbit, scroll to zoom, click any patient.

2. Unseen Patient Prediction



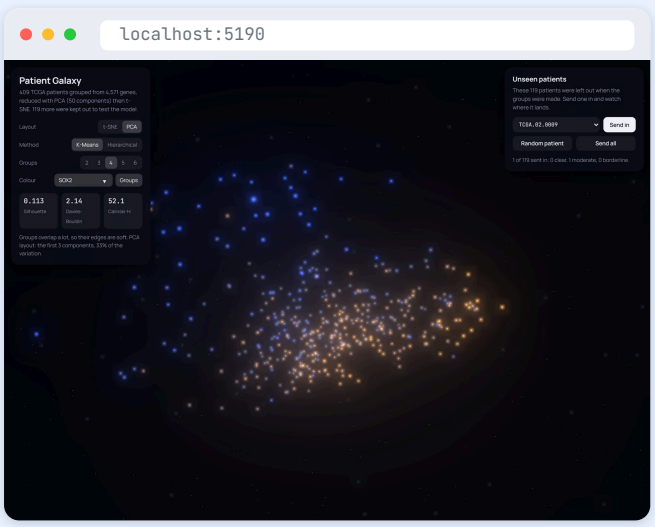
A held-out patient flies in and lands in **Group D**. Lines link the 15 most similar patients that voted.

3. Pipeline Dashboard

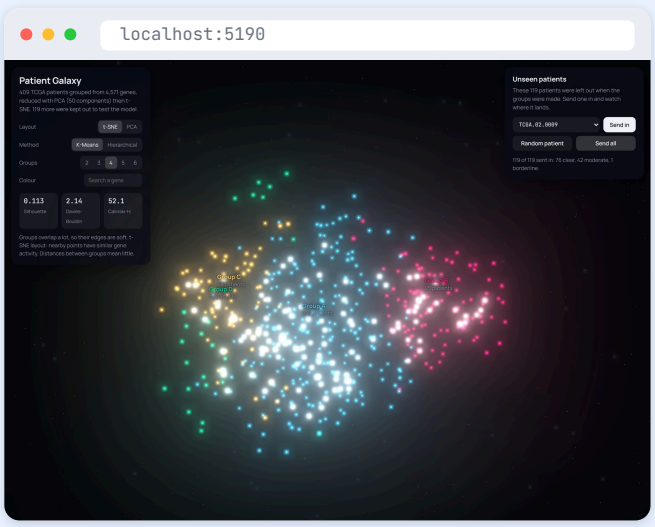


Set PCA, t-SNE, method and k, then read the **scores and maps**. A full run takes about 3 seconds.

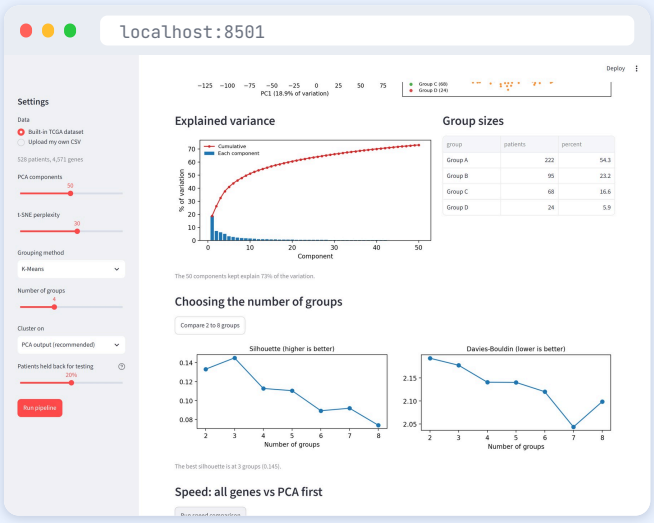
4. Additional Screens



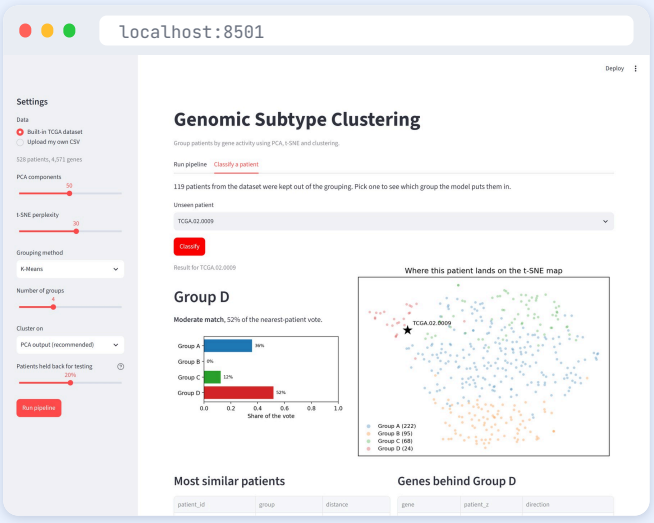
Colour by gene: SOX2 activity across all patients in the PCA layout.



Send all: 119 unseen patients land, 76 of them with a clear match.



Choosing k: scores for 2 to 8 groups and PCA explained variance.



Classify a patient: vote share per group and position on the t-SNE map.

Key Front End Features

- Interactive 3D patient galaxy built with three.js and WebGL
- Smooth morph between the t-SNE and PCA layouts
- Live switch of method (K-Means, Hierarchical) and k from 2 to 6
- Colour every patient by any gene's activity
- Unseen patients fly in and land by a nearest-patient vote
- Patient cards with similar patients and group marker genes
- Streamlit pipeline tool with live scores and speed tests
- CSV downloads of groups, metrics and predictions

Built with

Python · scikit-learn · pandas · Streamlit · three.js · WebGL · matplotlib

Runs on real TCGA data: 528 patients, with 119 kept out to test the model.